

Operating Systems for Intelligent Systems (ENGS2306)

Assignment 4

Process Exploration & Management

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1. Explore :

htop

Explanation: Shows a live, colorful dashboard of running processes.

```
deema@deema-al-maqadma: ~  
0[||||] 5.7% Tasks: 105, 228 thr; 1 running  
1[|] 1.3% Load average: 1.51 1.10 0.47  
Mem[||||||||||||||||] 817M/3.78G Uptime: 00:02:52  
Swp[|] 0K/2.09G  
  
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command  
2130 deema 20 0 11200 4608 3456 R 6.4 0.1 0:02.03 htop  
483 systemd-o 20 0 14836 6784 6016 S 0.6 0.2 0:00.73 /lib/systemd/systemd-oomd  
1608 deema 20 0 3859M 311M 130M S 0.6 8.1 0:20.17 /usr/bin/gnome-shell  
1622 deema 20 0 3859M 311M 130M S 0.6 8.1 0:04.36 /usr/bin/gnome-shell  
1623 deema 20 0 3859M 311M 130M S 0.6 8.1 0:04.15 /usr/bin/gnome-shell  
1 root 20 0 162M 11452 8124 S 0.0 0.3 0:04.68 /sbin/init auto noprompt splash  
338 root 19 -1 40084 17408 16128 S 0.0 0.4 0:01.18 /lib/systemd/systemd-journald  
381 root 20 0 27432 7296 4608 S 0.0 0.2 0:00.96 /lib/systemd/systemd-udevd  
485 systemd-r 20 0 25804 14040 9600 S 0.0 0.4 0:00.67 /lib/systemd/systemd-resolved  
486 systemd-t 20 0 89388 7424 6528 S 0.0 0.2 0:00.26 /lib/systemd/systemd-timesyncd  
498 systemd-t 20 0 89388 7424 6528 S 0.0 0.2 0:00.01 /lib/systemd/systemd-timesyncd  
538 root 20 0 234M 7628 6732 S 0.0 0.2 0:00.29 /usr/libexec/accounts-daemon  
540 root 20 0 2816 1920 1792 S 0.0 0.0 0:00.03 /usr/sbin/acpid  
545 root 20 0 8376 2304 2176 S 0.0 0.1 0:00.00 /usr/sbin/anacron -d -q -s  
557 avahi 20 0 7632 4096 3712 S 0.0 0.1 0:00.11 avahi-daemon: running [deema-al-maqadma.  
559 root 20 0 9496 2816 2688 S 0.0 0.1 0:00.01 /usr/sbin/cron -f -P  
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice + F9Kill F10Quit
```

```
1608 deema 20 0 3860M 310M 130M S 1.3 8.0 0:25.87 /usr/bin/gnome-shell  
483 systemd-o 20 0 14836 6784 6016 S 0.7 0.2 0:01.06 /lib/systemd/systemd-oomd  
1622 deema 20 0 3860M 310M 130M S 0.7 8.0 0:06.13 /usr/bin/gnome-shell  
1623 deema 20 0 3860M 310M 130M S 0.7 8.0 0:06.13 /usr/bin/gnome-shell  
483 systemd-o 20 0 14836 6784 6016 S 2.0 0.2 0:01.24 /lib/systemd/systemd-oomd  
1608 deema 20 0 3860M 310M 130M S 2.0 8.0 0:29.39 /usr/bin/gnome-shell  
1622 deema 20 0 3860M 310M 130M S 0.7 8.0 0:07.48 /usr/bin/gnome-shell  
561 messagebu 20 0 11028 6400 3968 S 0.0 0.2 0:01.25 @dbus-daemon --system --address=systemd:  
563 root 20 0 727M 19256 15800 S 0.0 0.5 0:00.92 /usr/sbin/NetworkManager --no-daemon  
591 root 20 0 41036 21120 11776 S 0.0 0.5 0:00.46 /usr/bin/python3 /usr/bin/networkd-dispa
```

2. Create & Identify: Start a process that sleeps for 5 minutes (300seconds) in the background:

sleep 300 &

Explanation:

Runs a sleep process in the background. The terminal prints its PID.

```
deema@deema-al-maqadma: ~  
deema@deema-al-maqadma:~$ sleep 300 &  
[1] 2389  
deema@deema-al-maqadma:~$
```

3. Find the Process PID

ps -ef | grep sleep

Explanation:

Searches for the sleep process and retrieves its PID.

```
deema@deema-al-maqadma:~$ ps -ef | grep sleep  
deema 2389 2067 0 10:15 pts/0 00:00:00 sleep 300  
deema 2427 2067 0 10:18 pts/0 00:00:00 grep --color=auto sleep  
deema@deema-al-maqadma:~$
```

4. Change its Priority

renice 12 -p <PID>

Explanation:

Lowens its priority by setting a niceness of 12 (higher number = lower priority).

```
deema@deema-al-maqadma:~$ sleep 300 &  
[1] 2803  
deema@deema-al-maqadma:~$ renice 12 -p 2803  
2803 (process ID) old priority 0, new priority 12
```

5. Verify Niceness

ps -o pid,ni,comm -p <PID>

Explanation:

Displays the process PID, its niceness level, and command name for confirmation.

```
deema@deema-al-maqadma:~$ ps -o pid,ni,comm -p 2803
  PID  NI  COMMAND
 2803  12  sleep
```

6. Terminate the Process

kill <PID>

Explanation:

Politely asks the process to terminate using SIGTERM.

```
deema@deema-al-maqadma:~$ kill 2803
deema@deema-al-maqadma:~$
```

7. Verify It's Gone

ps -ef | grep sleep

Explanation:

Confirms that the sleep process has been successfully terminated.

```
deema@deema-al-maqadma:~$ ps -ef | grep sleep
deema      2815      2067  0 10:50 pts/0    00:00:00 grep --color=auto sleep
[1]+  Terminated                  sleep 300
deema@deema-al-maqadma:~$
```

9. Report Explanation :

Option 1: SIGTERM vs SIGKILL \

- SIGTERM (15): A graceful exit signal. The process is asked to finish cleanly—saving data and releasing resources.

- SIGKILL (9): A forced termination signal. No cleanup—used when a process is frozen or refusing to exit.

Real scenario: You try to end a sensor-logging process with kill. If it doesn't respond, use kill -9.

Option 2: What is a Zombie Process?

A zombie process is a completed process that still exists in the process table because its parent didn't collect its exit status (via wait()). It doesn't consume resources but indicates poor process management.

System cleanup: The parent should clean it up. If not, the system eventually reassigns it to init, which will.

"واخر دعواهم ان الحمد لله رب العالمين"